



SQL OF THE DAY

define best-selling

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THE PROBLEM

Find the best-selling item per month, where best-selling means the highest total amount paid. Output the month, item description, and total paid.

returns hide as sales...

Medium · PostgreSQL



THE SOLUTION

```
WITH raw AS (  
    SELECT EXTRACT(MONTH FROM invoicedate) AS month,  
           description,  
           unitprice * quantity AS total_paid  
    FROM online_retail  
    WHERE quantity > 0  
) , agg AS (  
    SELECT month, description,  
           SUM(total_paid) AS total_paid  
    FROM raw GROUP BY month, description  
) , ranked AS (  
    SELECT month, description, total_paid,  
           RANK() OVER (PARTITION BY month  
                        ORDER BY total_paid DESC) AS rnk  
    FROM agg  
)  
SELECT month, description, total_paid  
FROM ranked WHERE rnk = 1;
```





HOW IT WORKS

REMOVE RETURNS

WHERE quantity > 0 drops refunds stored as negative quantities, which can push a returned item to the top.

REVENUE PER ROW

unitprice * quantity, then SUM to revenue per item per month.

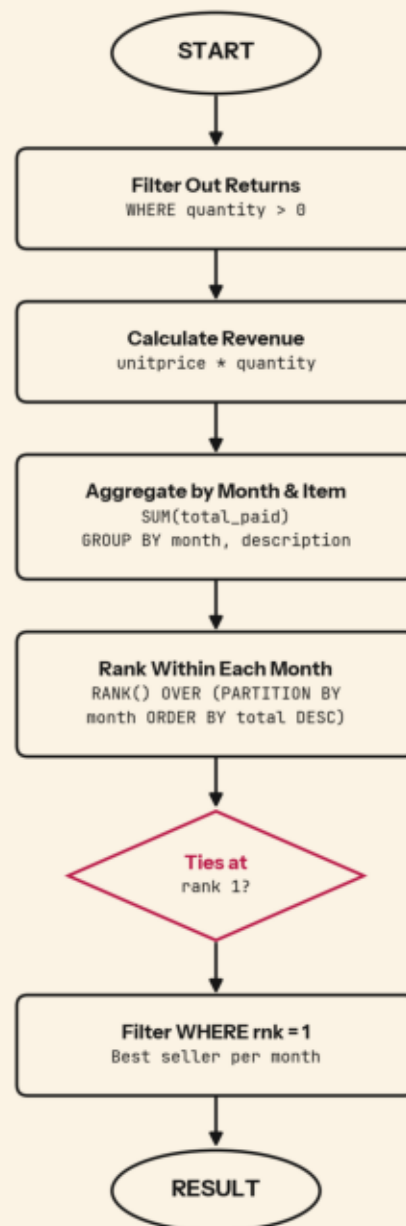
RANK PER MONTH

RANK() OVER (PARTITION BY month ORDER BY total DESC) keeps ties; WHERE rnk = 1 returns each month's winner.





THE LOGIC





WHY IT MATTERS

01

Inventory planning

Spot which items peak in which months and pre-order stock before demand arrives.

02

Margin

If an item already leads the month, a discount cuts margin without adding demand.

03

Seasonal strategy

Items that top the chart every year separate from one-time spikes, which shapes the core range.





KEY TAKEAWAYS

- 1 Define best-selling before writing the query. Revenue and units sold can crown different winners.
- 2 WHERE quantity > 0 removes refunds. Skip it and a heavily returned item can rank first by revenue.
- 3 RANK() keeps ties. ORDER BY ... LIMIT 1 picks one arbitrarily and looks complete when it is not.

follow for more SQL problems

questions? nivanrs@gmail.com

